

Homework (Habanero)

Q1. Solve the quadratic equation: $x^2 + 5x + 6 = 0$

- (A) $x = -2, x = -3$
- (B) $x = -1, x = -6$
- (C) $x = 2, x = 3$
- (D) $x = -2, x = 3$
- (E) $x = 1, x = 6$

Q2. Factor the quadratic expression: $x^2 - 7x + 12$

- (A) $(x - 3)(x - 4)$
- (B) $(x - 2)(x - 6)$
- (C) $(x + 3)(x - 4)$
- (D) $(x - 1)(x - 12)$
- (E) $(x + 1)(x + 12)$

Q3. The growth rate of a bacteria population is modeled by the equation $P(t) = 2t^2 + 5t + 1$, where t is time in hours. Find t when $P(t) = 15$.

- (A) $t = 1, t = 2$
- (B) $t = 1, t = 3$
- (C) $t = 2, t = 3$
- (D) $t = -1, t = 3$
- (E) $t = 1, t = -2$

Q4. Solve the quadratic equation: $x^2 - 4x - 3 = 0$

- (A) $x = -1, x = 3$
- (B) $x = 1, x = -3$
- (C) $x = 2, x = -1$
- (D) $x = -2, x = 1$
- (E) $x = 3, x = -1$

Q5. The dosage of a medication is given by the equation $D(t) = t^2 - 2t + 1$, where t is time in minutes. Find t when $D(t) = 0$.

- (A) $t = 1$
- (B) $t = 2$
- (C) $t = 3$
- (D) $t = 0$
- (E) $t = -1$

Q6. Factor the quadratic expression: $x^2 + 2x - 15$

- (A) $(x + 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 1)(x - 15)$
- (D) $(x - 1)(x + 15)$
- (E) $(x + 3)(x - 5)$

Q7. Solve the quadratic equation: $x^2 + x - 6 = 0$

- (A) $x = -3, x = 2$
- (B) $x = -2, x = 3$
- (C) $x = 1, x = 6$
- (D) $x = -1, x = 6$
- (E) $x = 3, x = -2$

Q8. The size of a cell is modeled by the equation $S(t) = t^2 + 4t + 3$, where t is time in hours. Find t when $S(t) = 16$.

- (A) $t = 1, t = 3$
- (B) $t = 2, t = 4$
- (C) $t = 1, t = 4$
- (D) $t = -1, t = 4$
- (E) $t = 2, t = -1$

Open Ended Questions (FRQ)

Q9. Solve the quadratic equation $x^2 - 6x + 8 = 0$ and verify the solutions.

Q10. The rate of a chemical reaction is modeled by the equation $R(t) = 2t^2 - 5t + 1$, where t is time in minutes. Find t when $R(t) = 5$ and explain your reasoning.

Chef's Tips

- Chef's Tips
- Encourage students to show their work and check their solutions.
- Remind students to use factoring and other methods to solve quadratic equations.